QUADRATIC FUNCTION

The Standard Form of a Quadratic Function is:

- The Function $f(x)=ax^2+bx+c$ where a, b, and c are constants and $a \neq 0$ is a quadratic function. Quadratic function in this form is said to be in standard form.

- The following are examples of quadratic functions.
  - $y=x^2$ a=1 b= 0 c =0
  - $f(x)=x^2+2x-5$ a=1 b= 2 c = -5
  - $g(x)=3x^2-4x$ a=3 b=-4 c= 0

continuation

- A quadratic function is any function that can be written in the form $f(x)= ax^2 + bx + c$, where $a \neq 0$.

EXAMPLE: Let $f(x) = (2x - 5)(x - 2)$. Show that f represents a quadratic function. Identify a, b, and c when the function is written in the form $f(x) = ax^2 + bx + c$.

Distribute each term in the first set of parentheses to each term in the second set of parentheses!$(2x - 5)(x - 2)$

$= 2x^2 - 4x - 5x + 10$

$= 2x2 - 9x + 10$

$a = 2, b = -9, c = 10$

continuation

- To find the value of x are called solutions or roots of the equation.
- The discriminant of the quadratic equation $ax^2 + bx + c = 0$ is given by $b^2 - 4ac$, and can be classify the solution of the equation as follow.

When $b^2 - 4ac$ is	Number of solutions
Perfect square	Two distinct rational solution
Positive non perfect square	Two distinct irrational solution
zero	One repeated rational solution
Negative number	Two distinct imaginary solution

- EXAMPLE: Use the determinant to determine the type of solutions of the quadratic equation.
  - $x^2 + 6x + 9 = 0$
  - $x^2 + 4x + 3 = 0$
  - $3x^2 + 5x + 5 = 0$
  - $x^2 + 4x + 2 = 0$